Overlay networks

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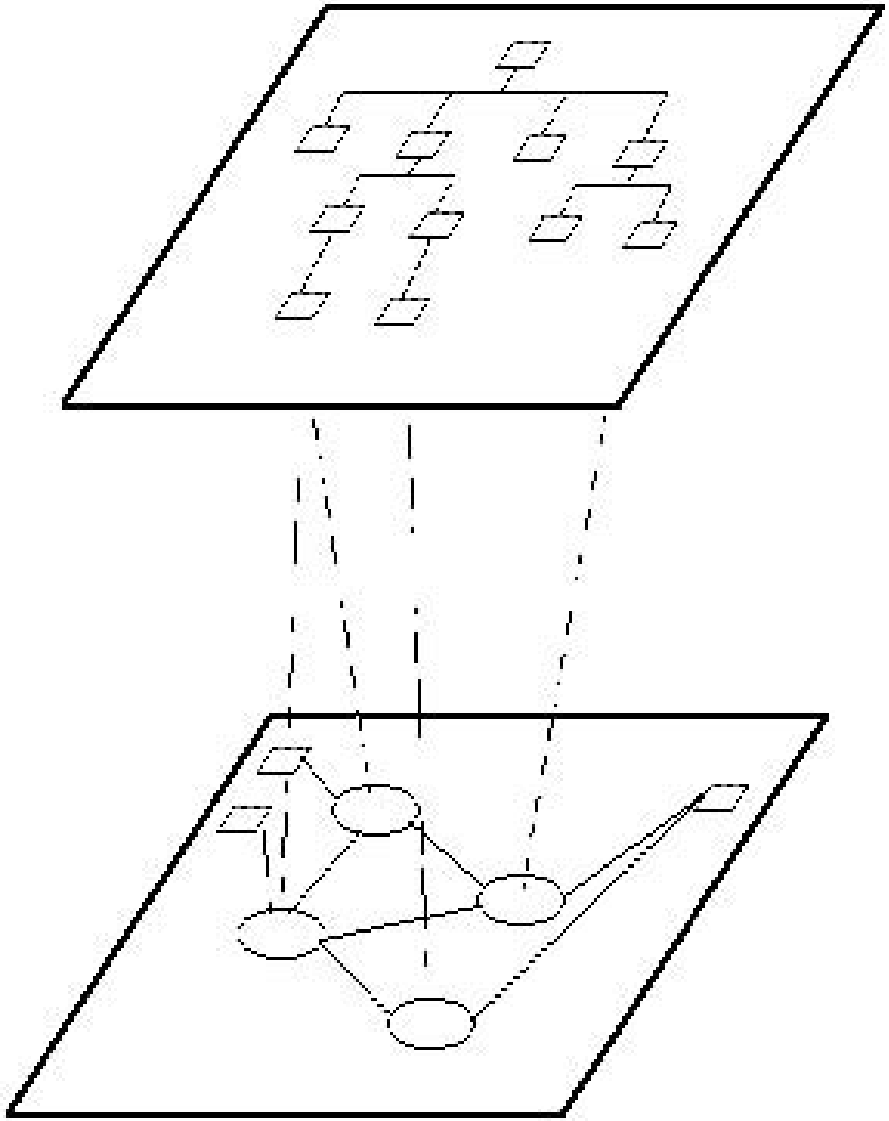
Outline

1 Introduction to overlay networks

Concept of overlay network I

- An overlay network is a computer network which is built on the top of another network.
- The aim is to implement a service that it is not available or it is not possible to offer in the underlying network.
- It generally provides new properties to the underlying network.
- Nodes in the overlay are connected via logical links.
- Logical (or virtual) links may consist of many physical links in the underlying network.

Concept of overlay network II



Concept of overlay network III

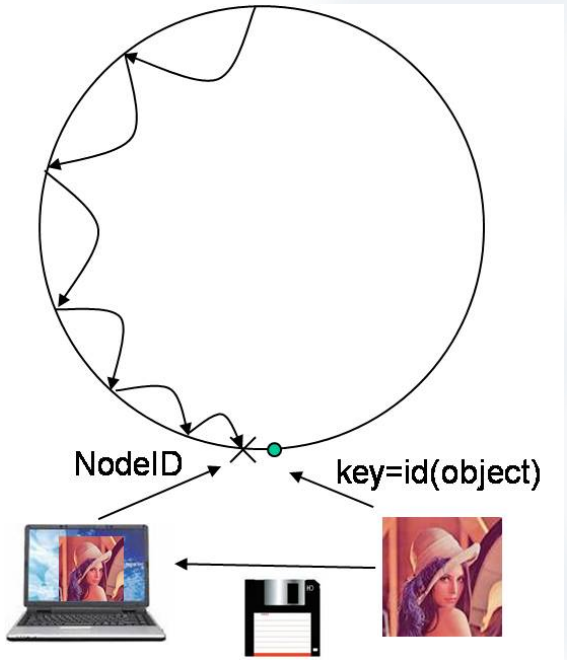
- Examples:
  - The Internet was originally built as an overlay upon telephone and (later) LAN networks.
  - Current distributed applications (client-server, p2p, cloud computing, etc) are overlay networks because their nodes run on top of the Internet.
  - With VoIP, the telephone network is increasingly turning into an overlay network built on top of the Internet.

Uses of overlays

- Improve routing.
- Provide new addressing schemes.
- Provide QoS.
- Provide security services.
- Multicasting, etc

Improving Routing

- Overlays permit to route messages to destinations not specified by IP addresses.
- The overlay has no control about how packets are routed in the underlying network, but it can control which overlay nodes are traversed to reach destination.
- Overlays can improve routing, for instance by means of distributed hash tables (DHT).



Assuring QoS

- Overlays can also guarantee a certain quality of service (QoS) in multimedia contents.
- Previous attempts:
  - IntServ (integrated services): based on reservations and RSVP (Resource Reservation Protocol). Fine-grained mechanism.
  - DiffServ (differentiated services): based on classifying and marking packets. Coarse-grained mechanism.
  - IP Multicast: group communications.
- They all require changes in all the routers.
- Tendency: deploy overlay networks in the end-systems (hosts) by means of software. We avoid changing routers, and the cooperation of ISPs.

Pros and Cons of overlays

- Pros:
  - Re-use of existing hardware/software/protocols.
  - Possibility to deploy different overlays over the same underlying network.
  - Not all the overlays offer the same properties/benefits.
  - Not all nodes participate in the (same) overlay.
- Cons:
  - Extra overhead (bandwidth, processing, storage, etc.).
  - Sometimes different layers provide redundant functions.

Some examples of overlays

- P2P (peer-to-peer).
- CDN (Content Delivery Networks)
- Cloud computing.
- Multimedia overlays.